Monitoring Task

Task Description:

Install Prometheus and Grafana on a Linux EC2 machine, connect Prometheus to Grafana, and create a dashboard to view metrics.

Provisioned EC2 instance with terraform

```
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\Kousick Devaraj\Desktop\DevOps\terraform\terraform-ec2-instance> cat .\main.tf
provider "aws" {
    region = "ap-south-1"
  }

  resource "aws_instance" "monitoring-server" {
    ami= "ami-01a00762f46d584a1"
    instance_type = "t2.medium"

    tags = {
        Name = "Observability-server"
    }
}
PS C:\Users\Kousick Devaraj\Desktop\DevOps\terraform\terraform-ec2-instance>
```

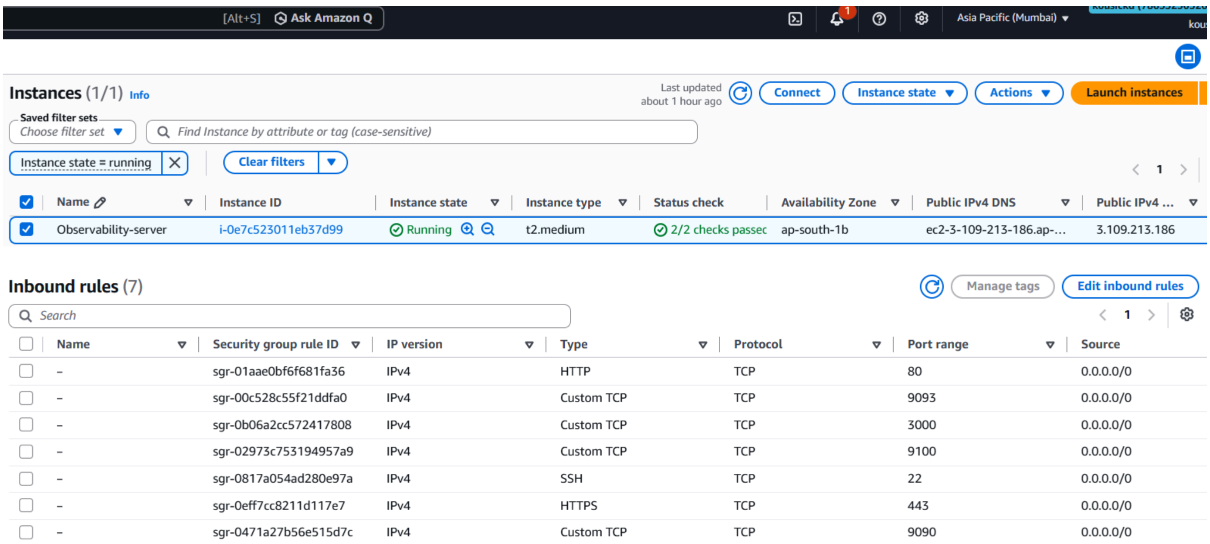
```
Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.monitoring-server: Creating...
aws_instance.monitoring-server: Still creating... [00m10s elapsed]
aws_instance.monitoring-server: Creation complete after 13s [id=i-0e7c523011eb37d99]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\Kousick Devaraj\Desktop\DevOps\terraform\terraform-ec2-instance>
```



Instances (1/1) Info

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
Observability-server	i-0e7c523011eb37d99	Running	t2.medium	2/2 checks passec	ap-south-1b	ec2-3-109-213-186.ap-...	3.109.213.186

Inbound rules (7)

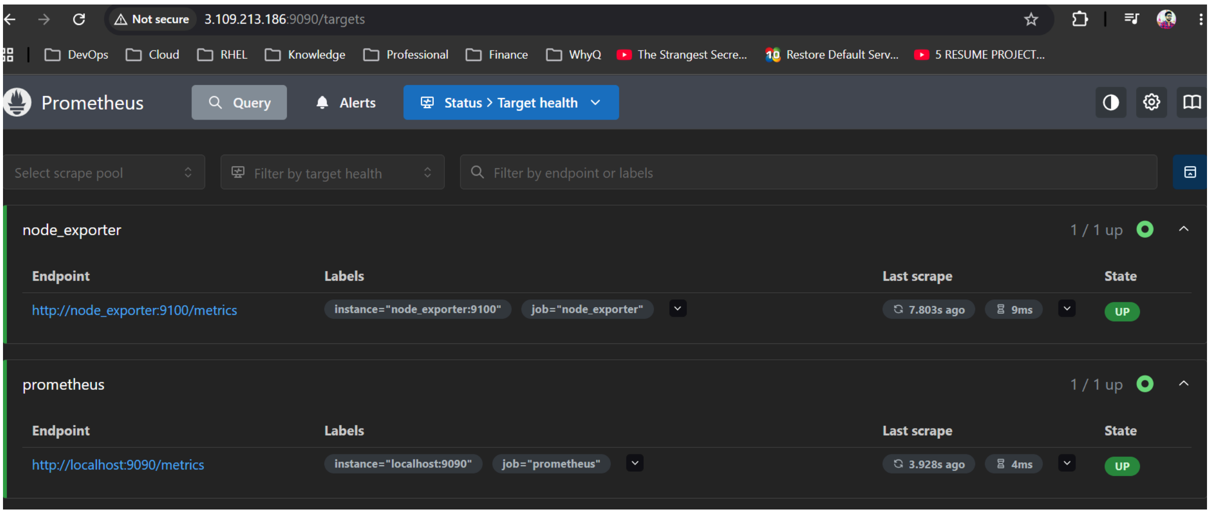
Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
–	sgr-01aae0bf6f681fa36	IPv4	HTTP	TCP	80	0.0.0.0/0
–	sgr-00c528c55f21ddfa0	IPv4	Custom TCP	TCP	9093	0.0.0.0/0
–	sgr-0b06a2cc572417808	IPv4	Custom TCP	TCP	3000	0.0.0.0/0
–	sgr-02973c753194957a9	IPv4	Custom TCP	TCP	9100	0.0.0.0/0
–	sgr-0817a054ad280e97a	IPv4	SSH	TCP	22	0.0.0.0/0
–	sgr-0eff7cc8211d117e7	IPv4	HTTPS	TCP	443	0.0.0.0/0
–	sgr-0471a27b56e515d7c	IPv4	Custom TCP	TCP	9090	0.0.0.0/0

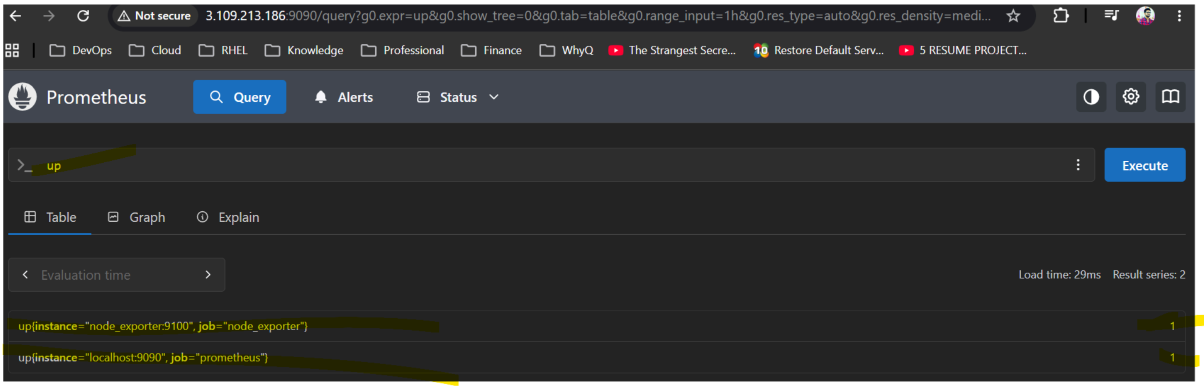
Installing prometheus and grafana using docker compose:

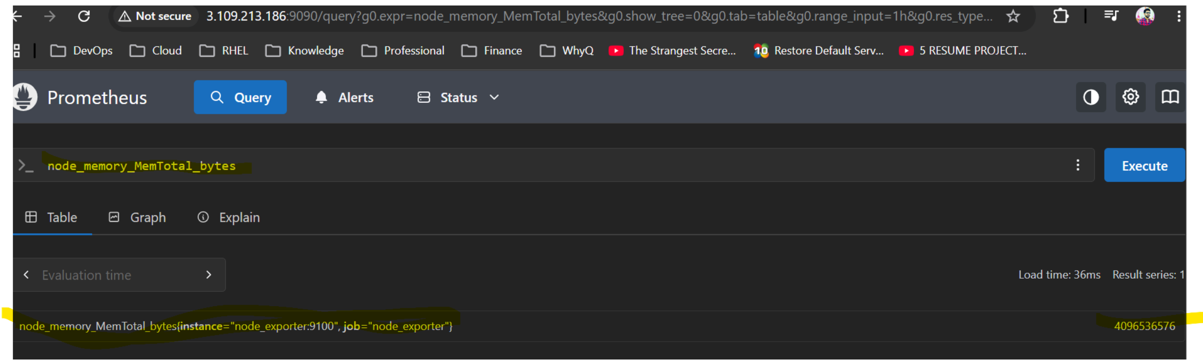
```
ubuntu@ip-172-31-2-73:~/monitoring$ ls -ltr
total 8
-rwxr-xr-x 1 ubuntu ubuntu 222 Aug 21 07:39 prometheus.yml
-rwxr-xr-x 1 ubuntu ubuntu 689 Aug 21 07:48 docker-compose.yml
ubuntu@ip-172-31-2-73:~/monitoring$
ubuntu@ip-172-31-2-73:~/monitoring$
```

```
ubuntu@ip-172-31-2-73:~/monitoring$ docker compose up -d
[+] Running 26/26
 ✓ grafana Pulled
   ✓ 23a073b63fe4 Pull complete
   ✓ 55afa1ecc21d Pull complete
   ✓ 6e5f17902521 Pull complete
   ✓ 530039403505 Pull complete
   ✓ adfbe35548cc Pull complete
   ✓ d563fee7e84d Pull complete
   ✓ 9baf42ea803d Pull complete
   ✓ 7e96b853aec8 Pull complete
   ✓ feefe83fec6c Pull complete
   ✓ 3395a07b0f7d Pull complete
   ✓ 4f4fb700ef54 Pull complete
 ✓ node_exporter Pulled
   ✓ f5ee56b8245c Pull complete
   ✓ c93f261857c3 Pull complete
   ✓ bef5f3176279 Pull complete
 ✓ prometheus Pulled
   ✓ 487858d29734 Pull complete
   ✓ da388505dad8 Pull complete
   ✓ e80cad428b4d Pull complete
   ✓ 200d6e325877 Pull complete
   ✓ 49c226752edd Pull complete
   ✓ 8ee344dffb17 Pull complete
   ✓ 7a5b1d5c3c45 Pull complete
   ✓ eb7a272b9937 Pull complete
   ✓ 0dc8fee0e5ee Pull complete
[+] Running 6/6
 ✓ Network monitoring_default           Created
 ✓ Volume monitoring_grafana_data       Created
 ✓ Volume monitoring_prometheus_data    Created
 ✓ Container prometheus                 Started
 ✓ Container grafana                    Started
 ✓ Container node_exporter              Started
ubuntu@ip-172-31-2-73:~/monitoring$
```

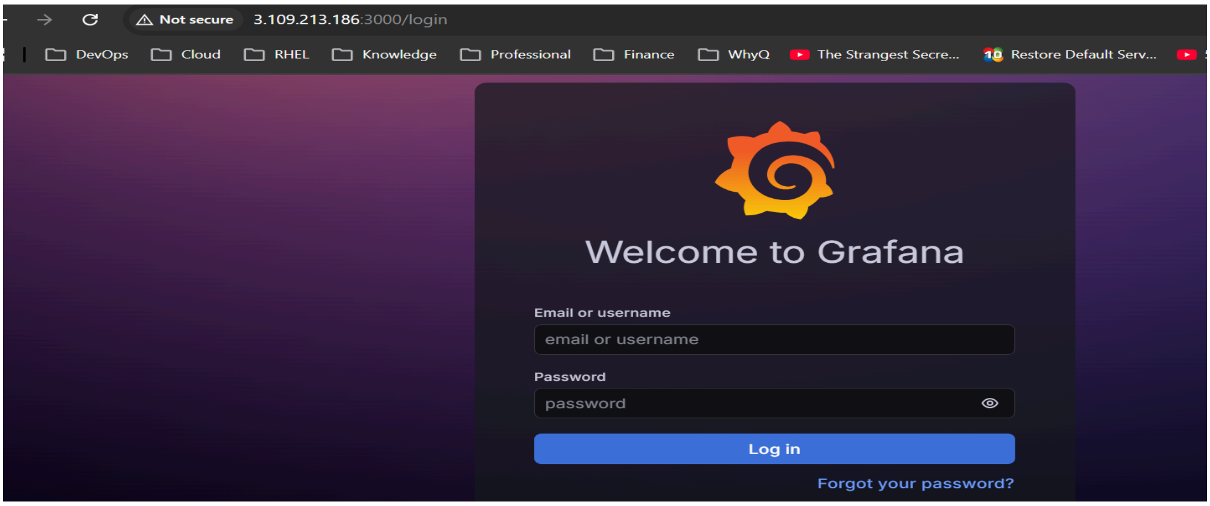
```
ubuntu@ip-172-31-2-73:~/monitoring$ docker compose ps
NAME            IMAGE                       COMMAND                  SERVICE         CREATED         STATUS         PORTS
grafana         grafana/grafana:latest      "/run.sh"                grafana         6 minutes ago   Up 6 minutes   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
node_exporter   prom/node-exporter:latest   "/bin/node_exporter"     node_exporter   6 minutes ago   Up 6 minutes   0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp
prometheus      prom/prometheus:latest      "/bin/prometheus --c…"   prometheus      6 minutes ago   Up 6 minutes   0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
ubuntu@ip-172-31-2-73:~/monitoring$ more prometheus.yml
```

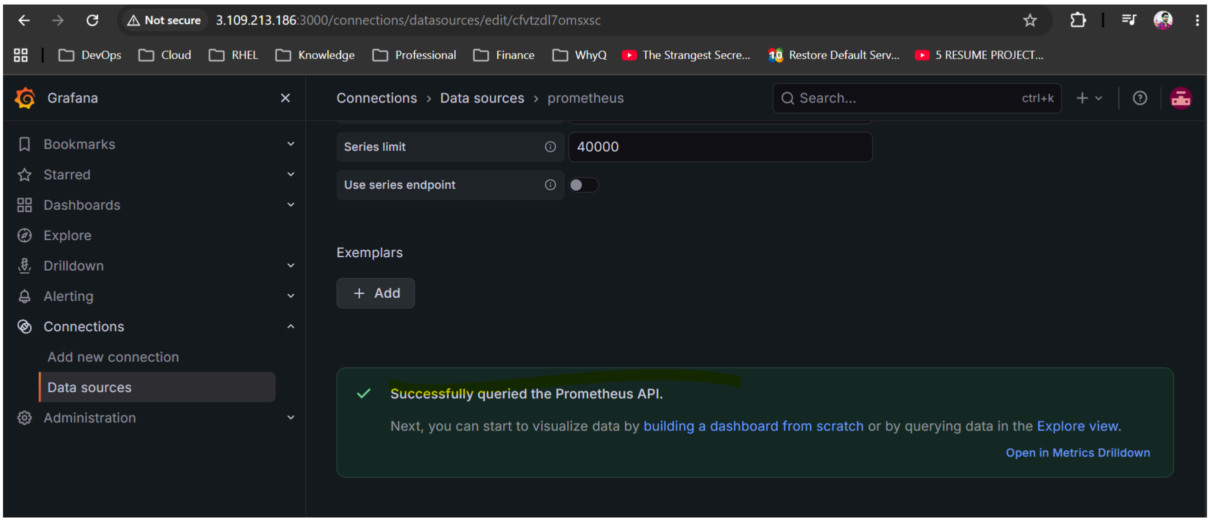






Grafana dashboard creation for monitoring prometheus server:


Not secure 3.109.213.186:3000/login
DevOps Cloud RHEL Knowledge Professional Finance WhyQ The Strangest Secre... Restore Default Serv...
Welcome to Grafana
Email or username
email or username
Password
password
Log in
Forgot your password?


Not secure 3.109.213.186:3000/connections/datasources/edit/cfvtzdl7omsxsc
DevOps Cloud RHEL Knowledge Professional Finance WhyQ The Strangest Secre... Restore Default Serv... 5 RESUME PROJECT...
Grafana
Connections › Data sources › prometheus
Search... ctrl+k
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Series limit 40000
Use series endpoint
Exemplars
Add
Successfully queried the Prometheus API.
Next, you can start to visualize data by building a dashboard from scratch or by querying data in the Explore view.
Open in Metrics Drilldown


Not secure 3.109.213.186:3000/d/adp7nd8/prometheus-monitoring?from=now-1h&to=now&timezone=browser&var-query0=
DevOps Cloud RHEL Knowledge Professional Finance WhyQ The Strangest Secre... Restore Default Serv... 5 RESUME PROJECT...
Grafana
Dashboards › Prometheus_monitoring
Search... ctrl+k
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Recently deleted
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
query0
Last 1 hour
Refresh
Save
Memory Utilization
CPU Utilization
{instance="node_exporter:9100", job="node_exporter"}
node_exporter:9100
Disk usage
73.1
73.1
73.1